

Advanced Programming in R

Initial Project Description

Students:

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Class time: Thursday 18:30
Preferred presentation week: 1st week of presentations

Project Proposal:

Poland Trip Planner in Your Pocket

Motivation.

We all love to travel — but most of the time we arrive in a new country with little idea of how its transport works, how long we should spend in each place, or how to fit everything we want to see into the limited days we have. Worse, we often pass right through towns with remarkable histories and never stop, simply because we didn't know they were there. All we usually have is a flight in, a flight out, and a list of places we dream of visiting.

This app aims to fix that for tourists coming to Poland.

User Input:

- Flight-in city and flight-out city
- Must-see cities
- Start date and finish date
- Preferred transport type — plane, bus, car, or train
- Travel style — scenic, fastest, or cheapest

How It Works:

1. The app calculates the most efficient order to visit all chosen cities, minimizing time spent travelling between them.

2. Time Allocation per City

The app uses an API (possibly OpenTripMap API, Wikidata API, Google Places API, or Gemini) to assess each city's size, historical significance, and cultural heritage, then proportionally distributes the available trip days across all destinations accordingly.

3. New Route Discovery

From a dataset of Polish cities, the app identifies the most important places that happen to lie along the route between destinations. The user can adjust a vicinity radius to control how

far off the path these suggestions reach — turning every journey into an opportunity for discovery. This only happens if they select filter that they are using car and they want "scenic" journey.

4. Transport Suggestions

For each leg of the trip, the app queries available flights and trains (possibly using APIs such as Amadeus Self-Service API , koleo.pl) and presents the best options filtered by the user's preferred transport type and travel style.

5. Carbon Footprint Dashboard

A side dashboard displays the minimum and maximum carbon footprint of the trip depending on which transport options the user selects, helping travellers make more informed and environmentally conscious choices.

Tools from the course we plan to use: Since we are a three students group - we will be using all 6 techniques covered during the advanced R course:

Writing advanced functions in R (including defensive programming), Object-oriented programming, C++, Code vectorisation and performance optimization, Shiny, R packages